



Tips for improving the identification of neoplastic visible lesions in Barrett's esophagus

Neoplastic-appearing visible lesions in Barrett's esophagus (BE) have a markedly high risk of harboring or developing invasive esophageal adenocarcinoma (EAC).¹ Therefore, recognition of neoplasia within the BE segment remains a critical component of early detection and treatment. Identification of visible lesions needs close inspection during endoscopy and their recognition as either flat or raised lesions with the use of high-definition white light endoscopy (WLE) or virtual chromoendoscopy (CE). Any superficial visible lesions identified during the examination should be categorized on the basis of the updated Paris classification as slightly elevated (0-IIa), completely flat (0-IIb), slightly depressed (0-IIc), excavated (0-III), or mixed pattern.

WHY IS RECOGNITION OF BE NEOPLASIA IMPORTANT?

The visible and potentially curable lesions, also called BE-related neoplasia (BERN), are defined as an abnormality of BE surface mucosa (by the Paris classification) or pattern indicative of dysplasia or cancer. There has been a significant increase in the prevalence of BERN lesions diagnosed at index upper endoscopy, from 5.1% (1990-1994) to 16.3% (2010-2019).² If these lesions are diagnosed at the early stage of high-grade dysplasia (HGD) or intramucosal adenocarcinoma (T1a), endoscopic treatments such as EMR or endoscopic submucosal dissection can provide excellent results, ranging from 85% to 99%. By contrast, failure to recognize BERN can lead to progression to an invasive lesion (T2 or higher), which can unfortunately reduce the survival to 10% (stages 3 and 4 EAC).

HOW OFTEN IS BE NEOPLASIA MISSED?

A critical question to ask is this: What is the miss rate of BERN in BE patients? An initial report by Tsoi et al³ from Australia noted that BE referral centers identified advanced BERN lesions in 27% of patients referred from the community with a diagnosis of low-grade dysplasia. In a similar study from the Netherlands, $\leq 23\%$ of patients with apparent flat BERN in community hospitals had higher grades of neoplasia after

repeated staging at an expert BE center.⁴ This provides an area of huge opportunity to improve methods to identify and manage visible lesions. This issue is now highlighted in a United States study that uses videos.⁵ Owing to the low rate of disease progression in BE to neoplasia ($<1\%$ per year), infrequency and unfamiliarity can be potential barriers to the recognition of subtle visible lesions in BE patients. Our understanding of BE diagnosis has evolved with the advent of the 1996 Los Angeles Classification, the development of the 2004 Prague C&M criteria, and the 2019 BERN program combined with efforts of the International Working Group for the Classification of

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esophagitis.⁶ The use of high-quality examinations, enhanced imaging modalities, education, quality metrics (eg, neoplasia detection rate [NDR]) and artificial intelligence (AI) can potentially further reduce the miss rate of BERN lesions. Although significant research and validation have occurred to standardize, adhere to guidelines, and improve the optical diagnosis of dysplasia, there are areas that need further improvement.

HOW TO IMPROVE BE NEOPLASIA RECOGNITION? TOP 5 TIPS FOR A HIGH-QUALITY EXAMINATION

Given the subtle nature of visible lesions in BE, a high-quality examination by an endoscopist with adequate experience in these abnormalities is essential (Table 1).

1. **Clean the esophagus.** First, with a combination of carbon dioxide insufflation and water jet, the

TABLE 1. Top 5 tips for a high-quality BE examination

5 Tips	Characteristics	Comments
Cleaning	Use of water with simethicone to clear saliva, bubbles, debris	It is recommended to use high-definition endoscopes; use of distal attachment device (cap) is extremely helpful
Inspection	High-quality inspection of entire esophageal segment with adequate inspection time	Longer time (1 min/cm) associated with higher rate of BERN detection
Virtual CE	Use of enhanced color imaging techniques such as NBI/BLI will identify mucosal and vascular patterns to detect and characterize BERN lesions	Studies have shown increased dysplasia detection (9%) with virtual CE compared with white light endoscopy alone
Education	Use of interactive web-based training and structured interventions to improve biopsy protocols and lesion recognition can improve neoplasia identification	Structured training and education can improve the neoplasia detection especially if incorporated early in the gastroenterology fellowship training
Measure your endoscopy unit's quality: NDR	Consider use of NDR as a quality metric to be incorporated as a quality metric and measured at the endoscopy unit level	Increase in NDR can decrease missed BERN lesions at the index endoscopy

BE, Barrett's esophagus; BERN, Barrett's esophagus-related neoplasia; BLI, blue-light imaging; CE, chromoendoscopy; NBI, narrow-band imaging; NDR, neoplasia detection rate.

esophageal mucosa should be cleaned and carefully inspected for the landmarks (diaphragmatic hiatus, gastroesophageal junction, and squamocolumnar junction).² A distal attachment cap could be used, especially in patients being referred with a diagnosis of neoplasia and those with a large hiatal hernia. All the mucus, saliva, and bubbles and any debris could be gently suctioned without creating mucosal trauma (Fig. 1).

2. Conduct a slow withdrawal with careful inspection. The entire esophageal segment should be inspected carefully, with adequate time spent during a gradual pullback after the stomach is deflated. A longer inspection time (at least 1 min/cm of BE) has shown a significantly higher rate of endoscopically suggestive lesions and overall rates of BERN (54.2% vs 13.3%; $P = .04$).⁷ Furthermore, a higher trend toward the detection of HGD/EAC was noted (40.2% vs 6.7%; $P = .06$).

3. Use virtual CE. The use of color (chromoendoscopy) during endoscopic inspection can improve the detection rates of BERN lesions. CE can be either virtual or dye based (acetic acid). Virtual CE incorporates methods to enhance inspection by using light filters (such as narrow-band imaging [NBI], blue-light imaging, and i-Scan). A meta-analysis of 25 studies including 2304 patients with BE reported a pooled sensitivity of 94% (95% confidence [CI], 83%-98%), negative predictive value of 98% (95% CI, 95%-98%), and specificity of 94% (95% CI, 81%-99%) for virtual CE.² Further, there was a 9% (95% CI, 4.1%-14%) increase in dysplasia detection with virtual CE compared with WLE.²

4. Improve education. Methods to improve the training of endoscopists, such as using an interactive web-based education tool, should be offered early in the career of GI fellows and could be incorporated into the educational curriculum.⁶ Even for gastroenterologists in practice, a

structured intensive educational intervention in BE examination has been shown to improve the use of the Seattle biopsy protocol and knowledge of BE patient care.⁸ Further, the use of centralized care involving tertiary centers that provide a multidisciplinary approach (with high-volume BE-treating endoscopists and access to experienced GI pathologists, creating centers of excellence) could improve BE management.

5. Follow the neoplasia detection rate (NDR) in your endoscopy unit. The NDR has recently been proposed as an important quality measure during endoscopic evaluation of BE. An NDR of approximately 4% (range 4-10%) has been used as a quality metric during upper endoscopy.⁹ Studies have shown that every 1% increase in the NDR can lead to a 3.5% decrease in the rate of later diagnosis of BERN lesions presumed to have been missed at the index endoscopy.²

ARTIFICIAL INTELLIGENCE: THE FUTURE

Given the high miss rate (up to one-quarter) of BERN lesions, AI is emerging as a promising adjunct tool in detecting subtle neoplastic lesions and improving sensitivity (>90%) and specificity (>80%).¹⁰

Machine learning algorithms that use deep learning and convolutional neural networks can assist in detecting BERN characterizing lesions, determining depth of invasion, and interpreting endoscopic images.¹⁰ Furthermore, AI systems could aid in reporting blind spots and areas not examined during upper endoscopy, and provide a grading score of the percentage of mucosa adequately inspected, including time spent, which can overall improve the quality of the BE endoscopic examination. The use of these technologies combined with real-time feedback and objective metrics can revolutionize the way we detect and manage BERN lesions.

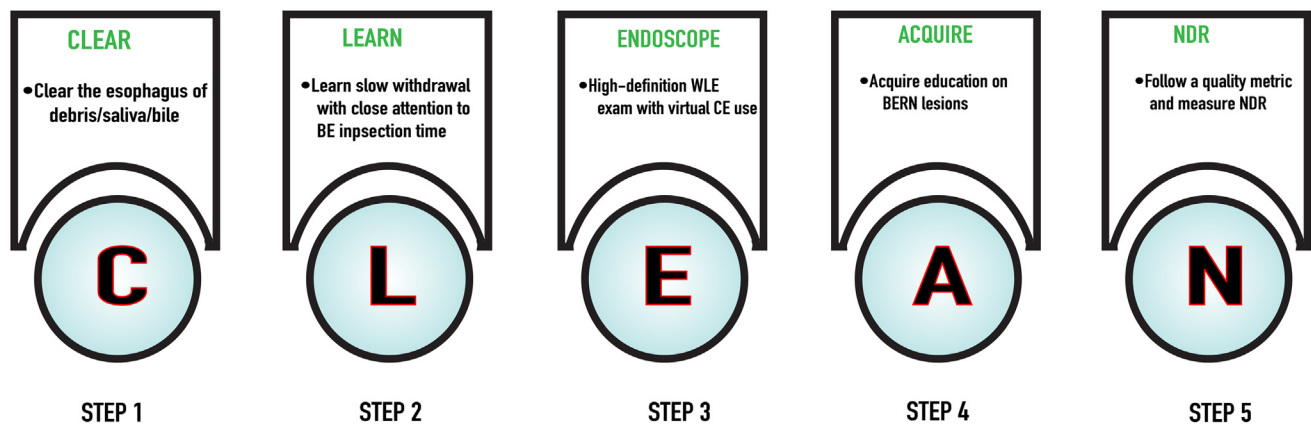


Figure 1. High-quality Barrett's esophagus examination. BE, Barrett's esophagus; BERN, Barrett's esophagus-related neoplasia; CE, chromoendoscopy; NDR, neoplasia detection rate; WLE, white-light endoscopy.

DISCLOSURE

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Abbreviations: AI, artificial intelligence; BE, Barrett's esophagus; BERN, Barrett's esophagus-related neoplasia; CE, chromoendoscopy; CLEAR, Clear; Learn, Endoscopy; Acquire, NDR; EAC, esophageal adenocarcinoma; HGD, high-grade dysplasia; NDR, neoplasia detection rate; WLE, white light endoscopy.

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